

Total No. of Questions :10]

SEAT No. :

P2615

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[5153] - 591

T.E. (Information Technology)

COMPUTER NETWORK TECHNOLOGY

(2012 Pattern) (Semester - I) (End-Sem.)

Time : 2½Hours]

[Max. Marks :70

Instructions to the candidates:

- 1) *Neat diagrams must be drawn wherever necessary.*
- 2) *Figures to the right indicate full marks.*
- 3) *Your answers will be valued as a whole.*
- 4) *Assume suitable data if necessary.*

Q1) a) Differentiate among circuit switching, packet switching and message switching with one example. **[6]**

b) Explain various transport layer services. **[4]**

OR

Q2) a) What is socket? Explain various client and server socket primitives. **[6]**

b) Write a short note on Quality of service. Parameters in Transport layer. **[4]**

Q3) a) A company is granted a site address 201.70.64.0. The company needs six subnets. Design the subnets. **[6]**

b) Lists the areas of network management and explain the necessity of any three. **[4]**

OR

Q4) a) What is DNS? What is server hierarchy? Explain domain name resolution process. **[6]**

b) Explain in detail how TCP provides flow control. **[4]**

P.T.O.

Q5) a) Explain the basic architecture of WLAN and discuss various components in it. [10]

b) Compare: Bluetooth and wireless LAN. [8]

OR

Q6) a) Explain Bluetooth features and architecture with suitable diagram. [10]

b) Explain frame format of 802.16. [8]

Q7) a) Explain following terms w.r.t. WSN: [8]

i) Data aggregation.

ii) Data diffusion.

iii) Data dissemination.

iv) Multicast.

b) What are different design issues of MAC protocol for WSN? [8]

OR

Q8) a) List different routing protocols used by WSN. Explain any one in detail. [8]

b) Explain Set up phase and steady state phase of LEACH protocol. [8]

Q9) a) Explain the tasks of address management in WSN. [8]

b) Describe DSDV Routing protocols. [8]

OR

Q10) Write short note on (any two): [16]

a) Internet of Things (IoT).

b) Software Defined Networking (SDN).

c) BYOD.

EEE